Z-TRANSFORM PROPERTIES AND PAIRS (WITH ROC)

Properties (ROC of X(z) is R: $r_1 < |z| < r_2$)

Property	Time domain	z-domain	ROC
Linearity	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$	at least $R_1 \cap R_2$
Shift (delay k)	$x(n-k)$	$z^{-k}\,X(z)$	R (0 or ∞ may be added or lost)
Scaling	$a^n\,x(n)$	$X(a^{-1}z)$	$\lvert a\rvert r_1 < \lvert z\rvert < \lvert a\rvert r_2$
Time reversal	$x(-n)$	$X(z^{-1})$	$1/r_2 < \lvert z\rvert < 1/r_1$
Differentiation	$n\cdot x(n)$	$-z\cdot$dX(z)/dz	R
Convolution	$x_1(n) * x_2(n)$	$X_1(z)\cdot X_2(z)$	at least $R_1 \cap R_2$ (bigger if a pole cancels)
Correlation	$r_{x_1x_2}(l) = x_1(l) * x_2(-l)$	$X_1(z)\cdot X_2(z^{-1})$	at least $R_1 \cap (1/R_2)$
Multiplication (windowing)	$x_1(n)\cdot x_2(n)$	$(1/2\pi\mathrm{j})\oint X_1(v)\,X_2(z/v)\,v^{-1}\mathrm{dv}$	see text (convolution of transforms)
Initial value (causal x)	$x(0)$	$\lim_{z\to\infty} X(z)$	—

Common transform pairs

x(n)	X(z)	ROC
$\delta(n)$	1	all z
$u(n)$	$1/(1-z^{-1})$	$\lvert z\rvert > 1$
$a^n u(n)$	$1/(1-az^{-1}) = z/(z-a)$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u(-n-1)$	$1/(1-az^{-1})$	$\lvert z\rvert < \lvert a\rvert$
$n\cdot u(n)$	$z^{-1}/(1-z^{-1})^2$	$\lvert z\rvert > 1$
$n\cdot a^n u(n)$	$az^{-1}/(1-az^{-1})^2$	$\lvert z\rvert > \lvert a\rvert$
$-n\cdot a^n u(-n-1)$	$az^{-1}/(1-az^{-1})^2$	$\lvert z\rvert < \lvert a\rvert$
$\sin(\omega_0 n)\,u(n)$	$\sin\omega_0\, z^{-1}/(1-2\cos\omega_0\, z^{-1}+z^{-2})$	$\lvert z\rvert > 1$
$\cos(\omega_0 n)\,u(n)$	$(1-\cos\omega_0\, z^{-1})/(1-2\cos\omega_0\, z^{-1}+z^{-2})$	$\lvert z\rvert > 1$
$a^n \sin(\omega_0 n)\,u(n)$	$a\sin\omega_0\, z^{-1}/(1-2a\cos\omega_0\, z^{-1}+a^2z^{-2})$	$\lvert z\rvert > \lvert a\rvert$
$a^n \cos(\omega_0 n)\,u(n)$	$(1-a\cos\omega_0\, z^{-1})/(1-2a\cos\omega_0\, z^{-1}+a^2z^{-2})$	$\lvert z\rvert > \lvert a\rvert$
finite sequence ($n_1 \dots n_2$)	$\sum x(n)\,z^{-n}$	all z except 0 ($n>0$ terms) and/or ∞ ($n<0$ terms)

QUICK RULES (TRAPS AND SHORTCUTS)

Sampling and the ADC

1. $f' = F/F_s$. Subtract the nearest whole number so $f' \in [-½, ½]$; alias $F' = |f'|\cdot F_s$. **sin flips sign if $f' < 0$**, cos does not. Tones with the same f' add ($3 + 10 = 13$).
2. D/A output: $F = f'\cdot F_s$ (only frequencies inside $\pm F_s/2$ come out). Out-of-band noise folds into the band, so filter **before** the ADC: pass edge B, stop edge $\boldsymbol{F_s - B}$, transition band = $F_s - 2B$ (≥ 0, so $F_s > 2B$ in practice).
3. Quantizer: $\Delta = R/(L-1)$ (from $L = R/\Delta + 1$; the +1 is *outside* the fraction); $L = 2^b$, $b = \lceil \log_2 L \rceil$. Quantize x: $k = \mathrm{round}((x - x_{\min})/\Delta)$, $x_q = x_{\min} + k\Delta$, code = k in binary, error ≤ $\Delta/2$. Sine range = 2A, $P = A^2/2$.

8 bits = 1 byte (kbit/s rate vs kB storage).

Shifts, folds, impulses, windows

1. Every operation **replaces every n** in the current formula: delay $k \to n-k$ (slides right); advance → $n+k$; fold → $-n$. Apply in the order stated: fold-then-delay-4 = $x(-(n-4)) = x(4-n)$; delay-then-fold = $x(-n-4)$ (that is the first result moved 8 left).
2. $\delta(n-k)$ sits at $n=k$. $u(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \dots$ ($n = 0, 1, 2, \dots$). $u(n+3) - u(n-4) = 1$ for $-3 \le n \le 3$ (7 samples). Fold/delay on a formula, e.g. $1 + n/3 \to 1 + (n-4)/3$.
3. Linearity zero-test only **rules out**: $y = 2x + 3$ and cos x fail; $y = x^2$ passes it but is still non-linear (T[2x] ≠ 2 T[x]). Delay = $x(n-k)$; a pure delay is time-invariant, $n\cdot x(n)$ and $x(2n)$ are not.

Convolution and impulse response

1. $y(n) = \sum x(k)\,h(n-k)$: for each n use every pair whose indices **add up to n**. Length $L_x + L_h - 1$, start = sum of the starts, check $\sum y = \sum x \cdot \sum h$. Equals polynomial multiplication in z^{-1}.
2. Finite geometric sum $\sum_{k=0}^{n} r^k = (r^{n+1}-1)/(r-1)$. Example: $(½)^n * (½)^n = (¼)^n(2^{n+1}-1)$. $h(n)$ for $x = \delta(n)$: at every step add the **feedback term** (coefficient × previous h) **and** the input terms; after the input ends $h(n) = p\cdot h(n-1)$.
3. Each input impulse launches a scaled, delayed copy of h: $x = -\delta(n+1) + 2\,\delta(n) + 3\,\delta(n-2) \Rightarrow y = -h(n+1) + 2\,h(n) + 3\,h(n-2)$. Exponent algebra: $a^{n+1} = a\cdot a^n$; combine only **like terms** (same base): $0.5\cdot 0.5^n - 0.5^n = -0.5\cdot 0.5^n$, but $0.5^n - 1^n = 0.5^n - 1$.

LCCDE, ZIR/ZSR and realizations

1. Solve for $y(n)$ to recurse: $y(n) = -\sum a_k y(n-k) + \sum b_m x(n-m)$. Normalize so $y(n)$ has coefficient 1; **a_k = coefficient next to $y(n-k)$ on the left**; drawn feedback gain = **$-a_k$**. Fill a table $n = 0, 1, 2$ (needs $y(-1), \dots$).
2. Homogeneous: λ^n, root of $\lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0$. $y + 0.5\,y(n-1) = x$ has $\lambda =$ **−0.5** (sign!). ZIR: $x = 0$ with the given $y(-1)$; ZSR: $y(-1) = 0$; total = ZI + ZS. Fit C with $y(0)$ from the recursion. Transient = decaying part, steady state = forced part ($H(1)$ × input).
3. DF-I uses $M + N$ delays, DF-II uses $\max(M, N)$. Missing term ⇒ coefficient 0 and no tap.

Z-transform, poles and stability

1. Replace n by $(n-k)$ in time ⇒ z^{-k} in z. Multiply by z^N to get positive powers; the list $[1, a_1, a_2, a_3]$ becomes $\boldsymbol{z^3 + a_1 z^2 + a_2 z + a_3}$ (same order, a_3 is the constant). **$w = z^{-1}$** trick to factor: (1 − ½w)(1 − ¼w) etc.; try ±1, ±2, ±½ as roots, divide out.
2. Calculator polynomial mode: enter a, b, c, d for ax³ + bx² + cx + d, use 0 for a missing power. Complex poles come in conjugate pairs (same magnitude): **magnitude** decides growth, angle only sets frequency.
3. Causal ⇒ ROC outside the outermost pole; stable ⇔ unit circle in ROC ⇒ all $|p| < 1$; pole on the circle = marginal, not BIBO stable. **Quick tests** on monic $f(z)$: $f(1) < 0$ ⇒ real root > 1 (unstable); $(-1)^N f(-1) < 0$ ⇒ root < −1; $|a_N| \ge 1$ ⇒ unstable; $\sum|a_k| < 1$ ⇒ stable. If these pass, compute the roots; they cannot prove stability alone.

PART A · DERIVATIONS (HOW TO ARRIVE AT THE EQUATION)

LECTURE 1 · SAMPLING & QUANTIZATION

Sampling relation and aliasing

1. $x(n) = x_a(\mathrm{n}T) = A\cos(2\pi F\cdot \mathrm{n}T + \varphi) = A\cos(2\pi(F/F_s)n + \varphi)$, so **$f = F/F_s$**.
2. For integer k and integer n: $\cos(2\pi(f+k)n + \varphi) = \cos(2\pi \mathrm{f} n + \varphi)$. So f and $f + k$ give the **same samples**; only $-½ \le f \le ½$ is distinct.
3. Alias: $f' = f - k$ (k = nearest integer), **$F' = F - \mathrm{k}F_s$**. To avoid aliasing need $|F| < F_s/2$: **$F_s \ge 2F_{\max}$**.

SQNR = 6.02b + 1.76 dB

1. Sine of amplitude A: range 2A, $L = 2^b$ levels ⇒ $\Delta \approx 2A/2^b$. Signal power $P_x = A^2/2$.
2. Rounding error e is uniform on $(-\Delta/2, \Delta/2)$: $P_q = (1/\Delta)\int \mathrm{e}^2\,\mathrm{de} = \boldsymbol{\Delta^2/12} = A^2/(3\cdot 2^{2b})$.
3. SQNR = $P_x/P_q = (3/2)\cdot 2^{2b}$. In dB: $10\log_{10}1.5 + 20\,b\cdot\log_{10}2 =$ **$1.76 + 6.02\,b$**.

LECTURE 2 · DT SYSTEMS

The convolution sum

1. Any signal is a sum of shifted impulses: $x(n) = \sum_k x(k)\,\delta(n-k)$.
2. Time invariance: $\delta(n-k) \to h(n-k)$. Linearity: scale and add.
3. **$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k) = x(n) * h(n)$**. Substitute $k \to n-k$ to get $h * x$ (commutative). Causal h ($h = 0$ for $n < 0$) and causal x: limits 0 … n.

BIBO stability ⇔ Σ|h(n)| < ∞

1. If $|x| \le M_x$: $|y(n)| \le \sum|h(k)||x(n-k)| \le M_x\sum|h(k)|$. Finite sum ⇒ bounded output (sufficient).
2. Necessary: choose $x(n) = \mathrm{sgn}\,h(-n)$; then $y(0) = \sum|h(k)|$, which is unbounded if the sum is.

LECTURE 3 · LCCDE

Direct method (homogeneous + particular)

1. $y(n) = y_h(n) + y_p(n)$.
2. **Homogeneous** ($x = 0$): try $y_h = \lambda^n$ in $\sum a_k y(n-k) = 0$ ⇒ characteristic polynomial **$\lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0$**. Distinct roots: $\sum C_i\lambda_i^n$; repeated root of multiplicity m: $(C_1 + C_2 n + \dots)\lambda^n$.
3. **Particular:** same form as the input (constant → K; a^n → Ka^n, use nKa^n if a is a root; cos/sin → K_1cos + K_2sin). Substitute to find K.
4. **Constants** C_i from the initial conditions applied to the *total* y (not to y_h). ZIR: $x = 0$ with the given $y(-1)$, $y(-2)$. ZSR: zero initial conditions.

Direct Form II from H(z)

1. $H(z) = B(z)/A(z) = [1/A(z)]\cdot B(z)$: do the all-pole part first. Define $W(z) = X(z)/A(z) \Rightarrow$ **$w(n) = x(n) - a_1 w(n-1) - \dots - a_N w(n-N)$**.
2. Then $Y(z) = B(z)\,W(z) \Rightarrow$ **$y(n) = b_0 w(n) + b_1 w(n-1) + \dots + b_M w(n-M)$**. One shared delay line of $\max(M, N)$ delays.
3. Direct Form I does $B(z)$ first (M delays on x), then $1/A(z)$ (N delays on y): $M + N$ delays.

LECTURE 4 · Z-TRANSFORM

Z{aⁿu(n)} and Z{−aⁿu(−n−1)}, with ROC

1. Causal: $X(z) = \sum_{n=0}^{\infty}(az^{-1})^n =$ **$1/(1 - az^{-1}) = z/(z-a)$** when $|az^{-1}| < 1$, *i.e.* **$|z| > |a|$**.
2. Anti-causal: $-\sum_{n=-\infty}^{-1}(az^{-1})^n = -\sum_{m=1}^{\infty}(z/a)^m = -(z/a)/(1 - z/a) =$ **$1/(1 - az^{-1})$** when **$|z| < |a|$**.
3. Same algebra, different ROC ⇒ different signal. $X(z)$ alone does not determine $x(n)$.

Differentiation: Z{n·x(n)} = −z·dX/dz

1. dX/dz = $\sum(-n)\,x(n)\,z^{-n-1} \Rightarrow -z\cdot$dX/dz = $\sum n\cdot x(n)\,z^{-n}$.
2. Example: $X = 1/(1 - az^{-1}) \Rightarrow$ dX/dz = $-(1 - az^{-1})^{-2}\cdot(az^{-2}) \Rightarrow$ **$Z\{n\cdot a^n u(n)\} = az^{-1}/(1 - az^{-1})^2$**, $|z| > |a|$.

Partial-fraction coefficients and complex poles

1. $X(z) = \sum A_k/(1 - p_k z^{-1})$. Multiply both sides by $(1 - p_k z^{-1})$ and set $z = p_k$: **$A_k = [(1 - p_k z^{-1})X(z)]_{z=p_k}$**.
2. Repeated pole: write $A/(1 - \mathrm{p}z^{-1}) + B/(1 - \mathrm{p}z^{-1})^2$ and use $(1 - \mathrm{p}z^{-1})^{-2} \leftrightarrow (n+1)\,\mathrm{p}^n\,u(n)$ (or the differentiation pair).
3. Complex pair p, p^* ⇒ A, A* (real signal). With A = $|A|e^{j\alpha}$, $p = |p|e^{j\beta}$: **$A\,p^n + A^*(p^*)^n = 2|A||p|^n\cos(\beta n + \alpha)$**.

cos and sin pairs

1. cos $\omega_0 n = ½(e^{j\omega_0 n} + e^{-j\omega_0 n})$: two exponentials a^n with $a = e^{\pm j\omega_0}$.
2. $½[1/(1 - e^{j\omega_0}z^{-1}) + 1/(1 - e^{-j\omega_0}z^{-1})] =$ **$(1 - \cos\omega_0\,z^{-1})/(1 - 2\cos\omega_0\,z^{-1} + z^{-2})$**, $|z| > 1$.
3. sin $\omega_0 n = (1/2\mathrm{j})(e^{j\omega_0 n} - e^{-j\omega_0 n}) \Rightarrow$ **$\sin\omega_0\,z^{-1}/(1 - 2\cos\omega_0\,z^{-1} + z^{-2})$**. For $r^n\cos/r^n\sin$ replace z^{-1} by rz^{-1} (scaling).

Shift, scaling, reversal, convolution

1. **Shift:** $\sum x(n-k)\,z^{-n}$, let $m = n-k$: $z^{-k}\sum x(m)\,z^{-m} = \boldsymbol{z^{-k}\,X(z)}$.
2. **Scaling:** $\sum a^n x(n)\,z^{-n} = \sum x(n)(a^{-1}z)^{-n} = \boldsymbol{X(a^{-1}z)}$; ROC becomes $|a|r_1 < |z| < |a|r_2$.
3. **Reversal:** $\sum x(-n)\,z^{-n}$, let $m = -n$: $\sum x(m)\,z^{m} = \boldsymbol{X(z^{-1})}$; ROC $1/r_2 < |z| < 1/r_1$.
4. **Convolution:** $\sum_n\sum_k x(k)\,h(n-k)\,z^{-n} = \sum x(k)\,z^{-k}\cdot\sum h(m)\,z^{-m}$ ($m = n-k$) = **$\boldsymbol{X(z)\,H(z)}$**.

H(z) from a difference equation, poles and zeros

1. Zero initial conditions, Z of both sides: $Y(z)(1+\Sigma a_k z^{-k}) = X(z)\,\Sigma\, b_m z^{-m} \Rightarrow \boldsymbol{H(z) = B(z)/A(z)}$.
2. Multiply top and bottom by z^N: $H(z) = b_0 z^{N-M}\Pi(z-z_k)/\Pi(z-p_k)$. Zeros z_k: $H=0$. Poles p_k: $H=\infty$ (extra poles/zeros at $z=0$ from $N-M$).
3. Improper ($M \geq N$): divide first, then partial fractions on the remainder. $h(n) = Z^{-1}\{H\}$ for the right ROC.

Stability and causality from the ROC

1. Causal ⇒ right-sided ⇒ ROC is outside the outermost pole ($|z| > r_{\max}$).
2. BIBO stable ⇔ $\Sigma|h| < \infty$ ⇔ $H(e^{j\omega})$ exists ⇔ **unit circle inside the ROC**.
3. Both ⇒ all poles satisfy $\boldsymbol{|p| < 1}$. A pole on the circle is marginal (an oscillator or accumulator), not BIBO stable.

One-sided Z and initial conditions

1. $X^+(z) = \Sigma_{n=0}^{\infty} x(n)\,z^{-n}$. For the delay: $\Sigma_{n=0}^{\infty} x(n-1)\,z^{-n} = x(-1) + z^{-1}X^+(z)$. Twice: $\boldsymbol{Z^+\{x(n-2)\} = z^{-2}X^+(z) + z^{-1}x(-1) + x(-2)}$.
2. Take Z^+ of the LCCDE, collect Y^+: $\boldsymbol{Y^+(z) = P(z)/A(z) + H(z)\,X(z)}$, where P comes only from $y(-1)$, $y(-2)$, ...
3. First term = zero-input response; second = zero-state response. Invert each term separately.

DESIGN DERIVATIONS

First-order smoother from a settling-time spec

1. $y(n) = a{\cdot}y(n-1) + (1-a)\,x(n)$, $0 < a < 1$; DC gain = $(1-a)/(1-a) = 1$.
2. Step response: $y(n) = 1 - a^{n+1}$. To reach 95 % at sample N: $a^{N+1} = 0.05 \Rightarrow \boldsymbol{a = 0.05^{1/(N+1)}}$.
3. Time constant $\tau \approx -1/\ln$ a samples (63 %); 95 % ≈ 3τ.

Moving-average null: zeros at the roots of unity

1. $H(z) = (1/M)\Sigma_{k=0}^{M-1} z^{-k} = (1/M)(1-z^{-M})/(1-z^{-1})$ (geometric sum).
2. Zeros: $z^M = 1$ except $z = 1 \Rightarrow \omega = 2\pi k/M$, $k = 1 \ldots M-1$. To null ω_0 choose $\boldsymbol{M = 2\pi/\omega_0}$ (first zero), which also nulls its multiples.

Digital oscillator (sine and cosine)

1. Want $y(n) = \sin(\omega_0 n)\,u(n)$ for $x = \delta(n)$: $H(z) = Y(z) = \boldsymbol{\sin\,\omega_0\,z^{-1}/(1 - 2\cos\,\omega_0\,z^{-1} + z^{-2})}$, $\omega_0 = 2\pi F/F_s$. Poles at $e^{\pm j\omega_0}$ (on the unit circle).
2. Difference equation: $y(n) = 2\cos\,\omega_0{\cdot}y(n-1) - y(n-2) + \sin\,\omega_0{\cdot}x(n-1)$. Cosine: numerator $1 - \cos\,\omega_0\,z^{-1}$.
3. No impulse? Preload the history: sine $y(-1) = -\sin\,\omega_0$, $y(-2) = -\sin 2\omega_0$; cosine $y(-1) = \cos\,\omega_0$, $y(-2) = \cos 2\omega_0$ (continue the sinusoid backwards).

Inverse system for an echo or ISI

1. Echo: $y(n) = x(n) + \alpha x(n-D) \Rightarrow H(z) = 1 + \alpha z^{-D}$. The equalizer is the inverse: $\boldsymbol{H_i(z) = 1/(1 + \alpha z^{-D})}$.
2. Difference equation: $\boldsymbol{x(n) = y(n) - \alpha x(n-D)}$. $h_i(n) = \Sigma(-\alpha)^k\delta(n - kD)$.
3. Causal and stable only if the poles ($|z|^D = |\alpha|$) lie inside the unit circle: $\boldsymbol{|\alpha| < 1}$.

Notch filter (remove ω₀, leave the rest)

1. Zeros on the circle at $e^{\pm j\omega_0}$ for an exact null: numerator $\boldsymbol{1 - 2\cos\,\omega_0\,z^{-1} + z^{-2}}$.
2. Poles at $r{\cdot}e^{\pm j\omega_0}$, just inside the circle ($r \approx 0.9$–0.99), cancel the zeros away from ω_0: denominator $\boldsymbol{1 - 2r\cos\,\omega_0\,z^{-1} + r^2z^{-2}}$.
3. Bandwidth ≈ $2(1-r)$ rad/sample. DC gain = $H(1) = (2 - 2\cos\,\omega_0)/(1 - 2r\cos\,\omega_0 + r^2)$. Closer $r \to 1$: narrower notch, longer ring-down.

ADC and quantizer sizing

1. Sampling: $F_s \geq 2\,F_{\max}$ (plus a guard band if the anti-alias filter is not ideal).
2. From an SQNR spec: $b \geq (\mathrm{SQNR_{dB}} - 1.76)/6.02$, round up. From a resolution: L = range/Δ + 1 ⇒ $b = \lceil \log_2 L \rceil$.
3. Bit rate = $b{\cdot}F_s$. Memory/delay of D samples costs $D{\cdot}b$ bits; D = delay$\cdot F_s$.

PART B · GRAPHING RULES

Stem plots: shifting, folding, scaling

1. Write the new signal as $x(\text{argument})$. Make a two-row table: n and argument, then read $x(\text{argument})$ at each n.
2. Delay $x(n-k)$: shift right by k. Fold $x(-n)$: mirror about $n=0$. Fold-then-delay $x(-(n-k)) = x(k-n)$: mirror, then place the old $n=0$ sample at $n=k$.
3. $x(2n)$ keeps every second sample (compressed, information lost); $x(n/2)$ inserts zeros. Mark $n=0$ with an arrow and label the heights.

Pole-zero plot with ROC

1. Draw the z-plane with the unit circle. Poles = $\boldsymbol{\times}$, zeros = $\boldsymbol{o}$ (write multiplicity). Remember extra poles/zeros at $z=0$ (or ∞) from $N-M$.
2. ROC is a circle, outside the circle, or a ring, bounded by poles. **Causal**: shade outside the outermost pole. **Anti-causal**: shade inside the innermost pole. **Two-sided**: the ring between two poles.
3. **No pole is ever inside the ROC.** Stable ⇔ the unit circle lies in the ROC. Finite sequences: all z except 0 and/or ∞.

Magnitude response from poles and zeros

1. Evaluate on the unit circle: $\boldsymbol{|H(e^{j\omega})| = |b_0| \cdot \Pi(\text{distance from } e^{j\omega} \text{ to each zero}) \,/\, \Pi(\text{distance to each pole})}$.
2. $\omega = 0$ is the point $z = 1$; $\omega = \pi$ is $z = -1$. Walking counter-clockwise: a pole near the circle at angle θ gives a **peak at $\boldsymbol{\omega \approx \theta}$**; a zero on the circle at θ gives a **null at $\boldsymbol{\omega = \theta}$**.
3. Notch: null at ω_0 with flat ≈ 1 elsewhere. Moving average: nulls at multiples of $2\pi/M$, lowpass-like with lobes. Oscillator: infinite peak (pole on the circle).

Aliasing picture (frequency axis)

1. Draw an axis in Hz with ticks at 0, $F_s/2$, F_s, $3F_s/2$ ... and fold it like a zig-zag at each multiple of $F_s/2$.
2. A tone at F lands at the same height on the first strip [0, $F_s/2$]: that is $F_{\text{alias}} = |F - \mathrm{k}F_s|$.
3. Sketch the samples: mark the sampling instants nT on the original sinusoid and connect them with the slower alias.

Impulse-response shape from the pole

1. Pole radius = decay: $|p| < 1$ decays, = 1 constant/oscillating, > 1 grows. Pole angle = oscillation: θ rad/sample ⇒ $F = \theta F_s/2\pi$.
2. Real positive pole: monotone. Real negative pole: alternates sign every sample. Complex pair: decaying cosine. Double pole: $n{\cdot}p^n$ (rises, peaks, decays).
3. First-order a^n: time constant $\tau = -1/\ln$ a; the envelope is 36.8 % of its start at $n = \tau$.

Block diagrams (DF-I, DF-II, FIR)

1. **FIR**: tapped delay line z^{-1} chain, a multiplier b_k per tap, one adder on the right.
2. **DF-II**: one delay chain $w(n-k)$. Left side adders form $w(n) = x(n) - \Sigma a_k w(n-k)$ (drawn gain = $-a_k$); right side forms $y(n) = \Sigma\, b_m w(n-m)$. Delays = $\max(M, N)$.
3. **DF-I**: input delays (b taps) feed an adder, then output delays (–a taps). Delays = $M + N$. Label every node, e.g. $x(n)$, $x(n-1)$, $y(n-1)$.

Sketching a convolution

1. Length $L_x + L_h - 1$, starts at (first index of x) + (first index of h). Total area $\Sigma y = \Sigma x \cdot \Sigma h$.
2. Two equal rectangles ⇒ triangle (peak = overlap length). Rectangle * exponential ⇒ rising then decaying. Causal exp * causal exp ⇒ starts at 0, tends to a smooth hump.
3. Per output point: fold h, slide it to n, multiply, sum. Draw the folded h under x and shade the overlap.

PART C · FORMULA SHEET

LECTURE 1 · SIGNALS & SAMPLING

Sampling relations

$$f = \frac{F}{F_s}$$

$x(n) = x_a(\mathrm{nT})$, $T = 1/F_s$. Distinct DT sinusoids: $-½ \leq f \leq ½$ ($-\pi \leq \omega \leq \pi$). Folding frequency $F_s/2$.

Quantization

$$\Delta = \frac{x_{\max} - x_{\min}}{L-1}, \quad L = 2^b, \quad P_q = \frac{\Delta^2}{12} \text{ (rounding)}$$

Truncation: mean-square error $\Delta^2/3$. Error $e_q(n) = x_q(n) - x(n)$, $|e_q| \leq \Delta/2$ for rounding.

SQNR

$$\mathrm{SQNR_{dB}} = 10\log_{10}\frac{P_x}{P_q}, \quad \text{full-scale sine: } 6.02b + 1.76$$

$P_x = (1/N)\Sigma x^2(n)$; a sinusoid has $A^2/2$.

Nyquist and aliasing

$$F_s \geq 2F_{\max} \qquad F_{\text{alias}} = F - \mathrm{k}F_s$$

Pick the integer k that puts F_{alias} in $[-F_s/2,\ F_s/2)$.

Quantize a sample

$$k = \mathrm{round}\left(\frac{x - x_{\min}}{\Delta}\right), \quad x_q = x_{\min} + k\Delta, \quad \Delta = \frac{x_{\max} - x_{\min}}{L-1}$$

Code = k in b-bit binary ($L = 2^b$). Error ≤ $\Delta/2$. Same as $L = (x_{\max} - x_{\min})/\Delta + 1$ solved for Δ; the +1 is outside the fraction.

Energy and average power

$$E = \Sigma|x(n)|^2, \quad P = \lim \frac{1}{2N+1}\Sigma_{-N}^{N}|x(n)|^2$$

Periodic: $P = (1/N)\Sigma$ over one period. Energy signal: $0 < E < \infty$ ($P = 0$). Power signal: $0 < P < \infty$ (E = ∞).

DT sinusoid periodicity

$$x(n+N) = x(n) \Leftrightarrow f = \frac{k}{N} \text{ rational}$$

Sum/product of sinusoids: N = lcm of the periods.

Even / odd parts

$$x_e = ½[x(n) + x(-n)], \quad x_o = ½[x(n) - x(-n)]$$

LECTURE 2 · DT SYSTEMS

Elementary signals

$$\delta(n), \quad u(n) = \Sigma_{k\geq 0}\delta(n-k), \quad r(n) = n{\cdot}u(n), \quad a^n u(n), \quad e^{j\omega n}$$

$\delta(n) = u(n) - u(n-1)$. Sifting: $x(n) = \Sigma_k\, x(k)\,\delta(n-k)$.

Convolution sum

$$y(n) = \Sigma_{k=-\infty}^{\infty} x(k)h(n-k) = x(n) * h(n)$$

Causal: $h(n) = 0$ for $n < 0$. Commutative, associative, distributive. Cascade: $h_1 * h_2$. Parallel: $h_1 + h_2$.

Linearity and time invariance

$$T[a_1x_1 + a_2x_2] = a_1T[x_1] + a_2T[x_2] \quad T[x(n-k)] = y(n-k)$$

Operations

delay: $x(n-k)$ advance: $x(n+k)$ fold: $x(-n)$ down-sample: $x(\mathrm{Mn})$

Block diagram elements: adder, multiplier, constant multiplier, unit delay z^{-1}, unit advance z.

BIBO stability

$$\Sigma_n\,|h(n)| < \infty$$

FIR: always stable. IIR: depends on pole locations.

Cross- and autocorrelation

$$r_{xy}(l) = \Sigma_n\, x(n)y(n-l), \quad r_{xy}(l) = r_{yx}(-l), \quad r_{xx}(l) = r_{xx}(-l)$$

$r_{xx}(0)$ = energy. $r_{yy} = r_{hh} * r_{xx}$; $r_{yx} = h * r_{xx}$. $r_{xy}(l) = x(l) * y(-l)$.

LECTURE 3 · LCCDE

General form

$$y(n) = -\Sigma_{k=1}^{N} a_k y(n-k) + \Sigma_{m=0}^{M} b_m x(n-m)$$

Non-recursive (FIR) if all $a_k = 0$. Recursive has feedback. Order = N.

Characteristic equation

$$\lambda^N + a_1\lambda^{N-1} + \ldots + a_N = 0$$

Roots = system poles. Distinct: $y_h = \Sigma\, C_k \lambda_k{}^n$. Repeated (multiplicity m): $(C_1 + C_2 n + \ldots + C_m n^{m-1})\lambda^n$.

Total response

$$y = y_h + y_p = y_{zi} + y_{zs}$$

$y_{zs} = h * x$ (relaxed). y_{zi} is the response to initial conditions with $x = 0$. y_p ↔ steady state, y_h ↔ transient (when stable).

Particular-solution forms

$$K \;\mid\; \mathrm{K a}^n \;\mid\; K_0 n^M + \ldots + K_M \;\mid\; K_1 \cos\omega_0 n + K_2 \sin\omega_0 n$$

Input: constant, a^n, polynomial n^M, sinusoid. ⚠ If the input form is already a homogeneous mode, multiply by n.

Impulse response of a recursive system

$$h(n) = y_{zs}(n) \text{ for } x = \delta(n)$$

Use $y(0) = b_0$, $y(1) = b_1 - a_1 b_0$ … to fit constants.

Direct Form II

$$w(n) = x(n) - \Sigma a_k w(n-k), \quad y(n) = \Sigma b_m w(n-m)$$

Memory = $\max(M, N)$ delays (DF-I uses $M + N$). ⚠ Drawn feedback gains are $-a_k$.

Second-order section

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) + b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

Building block for higher-order filters (cascade of SOS).

LECTURE 4 · Z-TRANSFORM

Definition

$$X(z) = \Sigma_{n=-\infty}^{\infty} x(n) z^{-n}$$

ROC = set of z with a finite sum. One-sided: $\Sigma_{n=0}^{\infty}$ (used with initial conditions).

Common pairs (ROC)

$$\delta(n) \leftrightarrow 1 \quad u(n) \leftrightarrow \frac{1}{1 - z^{-1}} \quad a^n u(n) \leftrightarrow \frac{1}{1 - a z^{-1}} \;\; |z|>|a|$$

$-a^n\, u(-n-1) \leftrightarrow 1/(1-az^{-1})$, $|z| < |a|$. $\mathrm{na}^n u(n) \leftrightarrow az^{-1}/(1-az^{-1})^2$. $\mathrm{na}^n u(n) \leftrightarrow az^{-1}/(1-az^{-1})^2$. $(n+1)a^n\, u(n) \leftrightarrow 1/(1-az^{-1})^2$.

Damped sinusoids

$$r^n \cos(\omega_0 n) u(n) \leftrightarrow \frac{1 - r\cos\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$$

$$r^n \sin(\omega_0 n) u(n) \leftrightarrow \frac{r\sin\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$$

$r = 1$ for undamped cos/sin. Poles at $re^{\pm j\,\omega_0}$.

Properties

$$x(n-k) \leftrightarrow z^{-k} X(z) \quad a^n x(n) \leftrightarrow X(a^{-1}z) \quad x(-n) \leftrightarrow X(z^{-1})$$

$$n \cdot x(n) \leftrightarrow -z\frac{\mathrm{dX}}{\mathrm{dz}} \quad x_1 * x_2 \leftrightarrow X_1 X_2$$

Also linearity, correlation $r_{x_0 x_2} \leftrightarrow X_1(z)\, X_2(z^{-1})$, windowing (product ↔ convolution of transforms). ⚠ Shift moves the whole expression (including u).

Initial value (causal)

$$x(0) = \lim_{z\to\infty} X(z)$$

System function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \ldots + b_M z^{-M}}{1 + a_1 z^{-1} + \ldots + a_N z^{-N}} = b_0 z^{N-M} \frac{\Pi(z - z_k)}{\Pi(z - p_k)}$$

Zeros z_k: $H = 0$. Poles p_k: $H = \infty$. FIR: all poles at 0. Purely recursive (all-pole): $b_1 \ldots b_M = 0$.

Stability and ROC

$$\text{BIBO stable} \Leftrightarrow \text{ROC} \ni |z| = 1 \quad \text{causal} \Leftrightarrow |z| > r_{\max} \quad \text{causal \& stable} \Leftrightarrow \text{all } |p_k| < 1$$

Partial-fraction residues

$$A_k = (1 - p_k z^{-1}) X(z)\big|_{z=p_k}$$

Conjugate poles ⇒ conjugate residues: $x(n) = 2|A||p|^n \cos(n\theta + \angle A)\, u(n)$. Repeated poles: differentiation property, $z/(z-p)^2 \leftrightarrow \mathrm{np}^{n-1} u(n)$.

One-sided shift

$$Z^+\{y(n-1)\} = z^{-1} Y(z) + y(-1)$$

$$Z^+\{y(n-2)\} = z^{-2} Y(z) + z^{-1} y(-1) + y(-2)$$

$Y(z)$ = ZIR(z) + $H(z)\, X(z)$. ZIR numerator comes only from initial conditions.

PART D · RECIPES

LECTURE 1 · SAMPLING & QUANTIZATION

Aliasing / what DT signal do I get?

1. Read each frequency F (Hz) from the argument: $\Omega = 2\pi F$.
2. $f = F/F_s$. If $|f| \le ½$ you are done: $x(n) = A\cos(2\pi\mathrm{fn} + \varphi)$.
3. Otherwise subtract the nearest integer: $f' = f - k$. The alias frequency is $F' = f' \cdot F_s$.
4. Reconstruction (ideal D/A) outputs only $|F| \le F_s/2$, so you get the alias, not the original.

cos is even (alias keeps its sign); sin is odd (alias flips sign when f' is negative). Add tones at equal frequency after aliasing.

Quantizer sizing and SQNR

1. $L = (x_{\max} - x_{\min})/\Delta + 1$ and $b = \lceil \log_2 L \rceil$ (or Δ = range/$(L-1)$ from b).
2. Rounding: $P_q = \Delta^2/12$. Truncation (mean-square): $\Delta^2/3$.
3. $P_x = A^2/2$ for a sinusoid, or $(1/N)\Sigma x^2$.
4. SQNR = $10\log(P_x/P_q)$; full-scale sine ⇒ $6.02\,b + 1.76$ dB.

Every extra bit = +6.02 dB. State which Δ convention (range/$(L-1)$ vs range/L) you used.

Quantize one sample (level, code, error)

1. $\Delta = (x_{\max} - x_{\min})/(L-1)$, with $L = 2^b$.
2. k = round$((x - x_{\min})/\Delta)$. Clip to $0 \ldots L-1$ if x is outside the range.
3. $x_q = x_{\min} + k\Delta$. Code = k in b-bit binary.
4. Error = $x_q - x$, and $|\text{error}| \le \Delta/2$ for rounding.

Example: range $-1 \ldots 1$, $L = 8$, $x = 0.30$: $\Delta = 2/7$, k = round$(1.3 \cdot 3.5)$ = round$(4.55) = 5$, $x_q = 0.429$, code 101, error 0.129 ($\le \Delta/2 = 0.143$). Truncation would use the floor instead of round.

Periodicity of a DT sinusoid

1. $f = \omega_0/2\pi$.
2. Rational (k/N in lowest terms)? Yes → fundamental period N. Irrational → not periodic.
3. Sum or product of sinusoids: N = lcm of the individual periods.

LECTURE 2 · DT SIGNALS & SYSTEMS

Shifting, folding and ordering

1. Write the final formula with n substituted into $x(\cdot)$: e.g. fold-then-delay-by-4 → $x(-(n-4)) = x(4-n)$.
2. Build a table of n against the argument of x, then read off $x(\text{argument})$.
3. Delay by k = shift right ($x(n-k)$); advance = shift left ($x(n+k)$).

Fold-then-delay ≠ delay-then-fold. Down-sampling $x(\mathrm{Mn})$ is time-variant.

Classify a system (static, linear, TI, causal, stable)

1. **Static?** Output depends only on $x(n)$ at the same n.
2. **Causal?** Every index in $x(\cdot)$ is ≤ n. Look at the worst case, e.g. $x(n^2+1)$, $x(4n)$, $x(-n)$ can reach the future.
3. **Linear?** Test $T[a_1 x_1 + a_2 x_2] = a_1 T[x_1] + a_2 T[x_2]$. Quick fail: zero in must give zero out.
4. **Time-invariant?** Feed $x(n-k)$: compare $T[x(n-k)]$ with $y(n-k)$. Different ⇒ time-variant (an explicit n multiplying the input, or n scaled inside $x(\cdot)$).
5. **BIBO stable?** Try a bounded input (e.g. $x = 1$): does y stay bounded?

Convolution table

1. Length of $y = L_x + L_h - 1$; first index = sum of first indices.
2. For each n: fold h, shift by n, multiply with x sample by sample, sum.
3. Check: $\Sigma y = \Sigma x \cdot \Sigma h$.
4. Alternatively, treat as polynomial multiplication of the coefficient lists.

Correlation

1. $r_{xy}(l) = \Sigma\, x(n)\, y(n-l)$: shift, multiply, sum (no fold).
2. $r_{xy}(l) = r_{yx}(-l)$; r_{xx} is even and peaks at $l = 0$ (= energy).
3. Through an LTI system: $r_{yy} = r_{hh} * r_{xx}$ and $r_{yx} = h * r_{xx}$.
4. To find a delay D buried in noise, find the lag where r_{yx} peaks.

LECTURE 3 · LCCDE

Direct method for y(n) = $y_h + y_p$

1. **Characteristic equation:** replace $y(n-k)$ by λ^{-k} and solve for λ.
2. **y_h:** distinct roots → $\Sigma C_k \lambda_k{}^n$; repeated root → $(C_1 + C_2 n)\lambda^n$; complex pair → $r^n(A\cos n\theta + B\sin n\theta)$.
3. **y_p:** same form as the input (K for a step, Ka^n for a^n, $K_1\cos + K_2\sin$ for sinusoids). If the input matches a root, multiply by n.
4. Compute $y(0)$, $y(1)$, … **by recursion** using the initial conditions.
5. Fit the C's to $y(0)$, $y(1)$ of the **total** solution $y_h + y_p$.

ZIR = response to initial conditions with $x = 0$. ZSR = response to the input with zero initial conditions (convolution with h). Total = ZIR + ZSR.

Recursion by hand

1. Rearrange: $y(n) = -\Sigma a_k y(n-k) + \Sigma b_m x(n-m)$.
2. Start at $n = 0$ with the given $y(-1)$, $y(-2)$, … (zero if relaxed).
3. For $h(n)$ set $x = \delta(n)$; the input terms vanish after $n = M$ and $h(n)$ follows the feedback alone.

Draw Direct Form I / II

1. **Normalize** so the coefficient of $y(n)$ is 1 (divide the whole equation).
2. DF-I: one delay line for x (M delays) and one for y (N delays); $y(n) = \Sigma\, b_m x(n-m) - \Sigma\, a_k y(n-k)$.
3. DF-II: $w(n) = x(n) - \Sigma a_k w(n-k)$, $y(n) = \Sigma b_m w(n-m)$; one shared line of $\max(M, N)$ delays.
4. Feedback multipliers in the drawing are $-a_k$; feedforward are b_m.
5. FIR = only the b taps (no feedback).

Check stability on the poles before realizing: roots of $\lambda^N + a_1\lambda^{N-1} + \ldots + a_N = 0$ must be inside the unit circle.

Read a block diagram

1. Label the output of every delay ($w(n-1)$, $w(n-2)$, …).
2. Write each adder as an equation exactly as drawn.
3. Eliminate the internal signal w to get the y–x relation.
4. Convert to the standard a_k/b_m form last.

LECTURE 4 · Z-TRANSFORM

Z-transform and ROC

1. Finite: write $X(z) = \Sigma x(n) z^{-n}$ term by term; ROC = whole plane minus 0 and/or ∞ depending on which powers appear.
2. $a^n u(n) \to 1/(1-az^{-1})$, $|z| > |a|$. $-a^n u(-n-1) \to$ same expression, $|z| < |a|$.
3. Sum of terms: ROC is the **intersection**; if empty, no Z-transform.
4. Use the property table (shift, scale, differentiate, convolve) instead of summing again.

ROC never contains a pole. Right-sided = outside, left-sided = inside, two-sided = ring.

Inverse Z (the decision tree)

1. **Improper?** (numerator degree ≥ denominator degree in z^{-1}) divide first: polynomial part → impulses.
2. **Factor** the denominator into $(1 - p_k z^{-1})$.
3. **Residues:** distinct p_k: $A_k = [(1 - p_k z^{-1})X(z)]$ at $z = p_k$. Repeated: expand $X(z)/z$ in $z/(z-p)^m$ terms. Complex pair: A and A*.
4. **Pick each term's side from the ROC:** pole inside the ROC's inner edge → right-sided $p^n u(n)$; pole outside → left-sided $-p^n u(-n-1)$.
5. Check $x(0)$ and $x(1)$ with long division.

From difference equation to H(z)

1. Replace $y(n-k) \to z^{-k} Y(z)$, $x(n-m) \to z^{-m} X(z)$.
2. $H(z) = Y/X = (b_0 + b_1 z^{-1} + \ldots)/(1 + a_1 z^{-1} + \ldots)$.
3. Multiply top and bottom by z^N to read poles/zeros as roots in z.
4. $h(n) = Z^{-1}\{H\}$; step response: invert $H(z)/(1-z^{-1})$.
5. Stable causal ⇔ all poles $|p| < 1$.

Nonzero initial conditions (one-sided Z)

1. Use $Z^+\{y(n-1)\} = z^{-1}\,Y + y(-1)$; $Z^+\{y(n-2)\} = z^{-2}\,Y + z^{-1}y(-1) + y(-2)$.
2. Collect: $Y(z)\cdot D(z) = N_{ic}(z) + N_{input}(z)X(z)$.
3. ZIR = N_{ic}/D; ZSR = $H(z)X(z)$.
4. Invert each part with the inverse-Z tree.

DESIGN PROBLEMS

Pole-zero placement

1. Convert the frequency to $\omega_0 = 2\pi F/F_s$ (rad/sample).
2. **To remove ω_0:** zeros at $e^{\pm j\omega_0}$ (numerator $1 - 2\cos\omega_0\, z^{-1} + z^{-2}$). Add poles at $r\cdot e^{\pm j\omega_0}$ with r close to 1 to keep neighbors untouched; width ≈ $2(1-r)$ rad.
3. **To pass ω_0 (resonate):** poles near $e^{\pm j\omega_0}$; $r = 1$ gives an oscillator (marginal).
4. **Smoothing:** real pole at $a \approx 1$; time constant ≈ −1/ln a samples.
5. Write $H(z)$ → difference equation → coefficients.
6. Verify: poles inside the circle, DC gain $H(1)$, and the frequency response at ω_0 and a neighbor.

Sizing a converter

1. $F_s = 2B \times (1 + \text{guard})$ and filter: pass B, stop $F_s - B$.
2. Bits: from the Δ spec (L = range/Δ + 1) or the SQNR spec ($b \geq (\text{SQNR} - 1.76)/6.02$).
3. Bit rate = $F_s \cdot b$.
4. Report the trade-offs (filter difficulty vs rate).

PART E · RED FLAGS

Red flags (the traps that cost points)

- **Aliasing:** subtract the integer *before* scaling by F_s; sin flips sign, cos does not. A DT sinusoid is periodic only if f is *rational*.
- **Fold/delay order:** fold-then-delay by k is $x(-(n-k))$; delay-then-fold is $x(-n-k)$. Different signals.
- **Linearity:** $y = 2x + 3$ is *not* linear (zero in ≠ zero out) but is time-invariant. $y = x^2$ is non-linear. $y = x(4n)$ is time-variant. $y = n\cdot x(n-1)$ is time-variant *and* unstable.
- **Convolution:** length = $L_x + L_h - 1$; first index = sum of first indices; $\Sigma y = \Sigma x \Sigma h$. Don't forget zero taps are still part of $h(n)$.
- **Recursive ≠ IIR.** A recursive running sum can be FIR (pole–zero cancellation); FIR ⇔ finite $h(n)$.
- **LCCDE:** fit C_1, C_2 with $y(0)$, $y(1)$ of the *total* solution; resonance (input = natural mode) ⇒ multiply y_p by n; repeated root ⇒ extra factor n.
- **Realization:** normalize the $y(n)$ coefficient to 1 first; drawn feedback gain is $-a_k$; DF-II uses $\max(M, N)$ delays, DF-I uses $M + N$.
- **Z-transform shift:** $a^n u(n-2) \neq a^{n-2} u(n-2)$ — rewrite before using z^{-2}. Time reversal flips the ROC to the inside.
- **ROC:** never contains a pole. $X(z)$ without ROC does not define $x(n)$. Sum of right- and left-sided pieces may have *empty* ROC (no transform).
- **Improper $X(z)$:** divide first; the polynomial part gives impulses at the start of $x(n)$.
- **Stability:** BIBO stable ⇔ unit circle in the ROC. Causal ⇒ all poles inside $|z| < 1$. A pole at 2 can give a stable system only if non-causal.
- **Pole angle ↔ frequency:** $\omega_0 = 2\pi F/F_s$. Radius ↔ decay; $r = 1$ ⇒ sustained oscillation (marginal); $r > 1$ ⇒ growth.

Numbers worth knowing cold

- SQNR(full-scale sine) = $6.02b + 1.76$ dB; every bit adds 6.02 dB; truncation is 6.02 dB worse than rounding (mean-square).
- 2cos 36° = 1.618 (golden ratio), sin 36° = 0.5878, cos 45° = sin 45° = 0.7071, cos 60° = 0.5, sin 60° = 0.866.
- $\Sigma_{k=0}{}^n\, a^k = (1 - a^{n+1})/(1 - a)$; $\Sigma_{k=0}{}^{\infty}\, a^k = 1/(1 - a)$ for $|a| < 1$.
- Step response of $1/(1 - az^{-1})$: $(1 - a^{n+1})/(1 - a)$; settles (95 %) in about $3/(-\ln a)$ samples.
- Moving average of length M: zeros at $\omega = 2\pi k/M$, DC gain 1, $h = 1/M$ each.